

Hybrid Model Predictive and Nonlinear Feedback Control for Chaos Suppression and MPPT Enhancement in DFIG-Based Wind Energy Systems

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Abstract—Doubly fed induction generator (DFIG)-based wind turbines, despite their widespread adoption, can exhibit chaotic dynamics under adverse conditions such as grid disturbances, turbulent wind profiles, and parameter deviations, leading to degraded power quality and compromised stability. This paper investigates the emergence of chaotic behavior in variable-speed DFIG systems through comprehensive nonlinear modeling and proposes an innovative hybrid control architecture combining Model Predictive Control (MPC) with nonlinear state-feedback compensation. The proposed controller effectively suppresses chaotic dynamics, achieves high-precision tracking of Maximum Power Point Tracking (MPPT)-derived reference trajectories, and demonstrates superior robustness against parameter uncertainties under realistic wind conditions. Simulation results validate the effectiveness of the hybrid control strategy in maintaining stable operation, minimizing tracking error, and ensuring compliance with stringent grid-code requirements, thereby enhancing the reliability and performance of DFIG-based wind energy systems.

Index Terms—Doubly-Fed Induction Generator (DFIG), Wind Energy Systems, Chaotic Dynamics, Nonlinear Dynamics, Maximum Power Point Tracking (MPPT), Hybrid Control, Model Predictive Control (MPC), Nonlinear Feedback Control, Robust Control.

I. INTRODUCTION

The escalating global population coupled with increasing electricity demand has accelerated the depletion of non-renewable energy resources at unprecedented rates [1]. This critical challenge necessitates the exploration and deployment of alternative, sustainable, and environmentally-friendly energy sources, particularly wind energy, to mitigate environmental degradation and address energy security concerns [2], [3].

Wind energy has emerged as a pivotal renewable energy technology in recent decades, experiencing substantial growth in both global investments and installed capacity [4]. Recent statistics indicate that global wind power capacity expanded by a record 117 GW in 2024, elevating the cumulative worldwide installed capacity to approximately 1,136 GW [5]. This remarkable expansion has been driven by supportive policy frameworks, technological innovations, and declining leveled cost of energy, positioning wind power as one of the fastest-growing renewable energy sources worldwide [6].

Among various wind turbine generator configurations, the doubly fed induction generator (DFIG) has become the dominant technology in large-scale onshore wind farms, primarily due to its partial-scale back-to-back power converter architecture that enables flexible control of active and reactive power with reduced converter ratings [7], [8]. This configuration allows the rotor-side converter (RSC) and grid-side converter (GSC) to independently regulate rotor currents and grid-injected currents, respectively, thereby achieving decoupled control of electromagnetic torque, stator active power, and reactive power over a wide operational speed range [9].

The economic advantage of requiring only 25–30% rated power converters compared to full-scale converter systems, combined with superior power quality and grid-support capabilities, has made DFIG the preferred choice for variable-speed wind energy conversion systems (WECS) [10].

Despite extensive research on DFIG-based WECS including dynamic modeling, vector control strategies, grid-fault ride-through capability, and ancillary services provision, recent studies have revealed that these systems can exhibit complex nonlinear dynamic behaviors under certain operating conditions [11]–[13]. The inherent nonlinearity of the induction machine, coupled with the interaction between mechanical and electrical subsystems and the coupling between stator and rotor flux linkages, can lead to oscillatory transients and, under severe conditions, bifurcations and quasi-chaotic responses [14], [15].

These phenomena become particularly pronounced under severe grid disturbances, highly turbulent wind speed profiles, parameter uncertainties (such as machine detuning), and converter saturation effects, all of which may degrade power quality, stress mechanical components, and compromise system stability [16], [17].

The emergence of chaotic behavior in DFIG systems manifests as irregular power oscillations, torque ripple, and complex trajectories in the state space, which are detrimental to both electrical and mechanical performance [18]. Such dynamics are especially critical in the context of stringent grid-code requirements for low-voltage ride-through (LVRT), frequency support, and reactive power support [19], [20].

Consequently, the analysis, mitigation, and control of nonlinear and potentially chaotic dynamics in DFIG-based wind turbines have become important research priorities for ensuring stable operation under wide-ranging operating scenarios and maximizing the contribution of wind energy to global electricity supply.

Motivated by these challenges, this paper investigates the control of a variable-speed wind turbine equipped with a DFIG that exhibits chaotic-like behavior induced by machine parameter deviations and adverse operating scenarios, which can severely affect system stability and grid-injected power quality. A comprehensive nonlinear mathematical model is first developed to capture the intricate coupling among rotor currents, electromagnetic torque, and mechanical speed, and to identify the conditions under which complex or chaotic regimes emerge when the system is subjected to significant disturbances, such as abrupt wind-speed variations or degraded parameter settings. Moreover, the system is subjected to a highly realistic wind-speed profile to evaluate its ability to maintain stability and drive its dynamics toward accurate wind-path tracking, as assessed through the tracking error. Building on this framework, an innovative hybrid control architecture is then proposed, combining Model Predictive Control (MPC) with nonlinear state-feedback compensation to suppress chaotic dynamics, enhance robustness to uncertainties, and achieve high-precision tracking of reference rotor current and speed trajectories derived from Maximum Power Point Tracking (MPPT) objectives.

II. WIND TURBINE MODELING

The aerodynamic power captured by a horizontal-axis wind turbine can be written as [21]–[23]

$$P_a = \frac{1}{2} \pi \rho R^2 C_p(\lambda) v^3 \quad (1)$$

where ρ is the air density, R is the rotor radius, v is the wind speed, and $C_p(\lambda)$ is the power coefficient, which depends on the tip-speed ratio λ .

The tip-speed ratio is defined by

$$\lambda = \frac{R\omega_r}{v} \quad (2)$$

with ω_r the rotor (mechanical) angular speed.

The corresponding aerodynamic torque applied to the shaft is obtained from

$$T_a = \frac{P_a}{\omega_r} = \frac{1}{2} \pi \rho R^2 C_p(\lambda) \frac{v^3}{\omega_r} \quad (3)$$

The rotational dynamics of the wind turbine-generator set are described by

$$J \frac{d\Omega_m}{dt} = T_a - T_e - D\Omega_m \quad (4)$$

where J is the total inertia, Ω_m is the mechanical rotor speed, T_e is the electromagnetic torque produced by the generator, and D denotes the viscous friction coefficient.

Relating mechanical and electrical rotor speeds by

$$\omega_r = n_p \Omega_m \quad (5)$$

with n_p the number of pole pairs, and using $\frac{d\Omega_m}{dt} = \frac{1}{n_p} \frac{d\omega_r}{dt}$, (4) can be rewritten in electrical speed form as

$$\frac{J}{n_p} \frac{d\omega_r}{dt} = T_a - T_e - \frac{D}{n_p} \omega_r \quad (6)$$

For a DFIG expressed in the stator dq reference frame, the electromagnetic torque is given by

$$T_e = \frac{3}{2} n_p (\psi_{ds} i_{qs} - \psi_{qs} i_{ds}) \quad (7)$$

where ψ_{ds}, ψ_{qs} are the stator flux components and i_{ds}, i_{qs} are the corresponding stator current components.

Substituting (7) into (6) yields the final electromechanical dynamic equation of the rotor:

$$\frac{J}{n_p} \frac{d\omega_r}{dt} = T_a - \frac{3}{2} n_p (\psi_{ds} i_{qs} - \psi_{qs} i_{ds}) - \frac{D}{n_p} \omega_r \quad (8)$$

III. CHAOTIC DOUBLY-FED INDUCTION GENERATOR

According to (8) and the stator-flux/rotor-current coupling of the DFIG, a third-order state equation with i_{dr} , i_{qr} , and ω_r as state variables can be obtained [24]:

$$\begin{aligned} \dot{i}_{dr} &= \left(-\frac{R_s L_m^2}{\sigma L_s^2 L_r} - \frac{R_r}{\sigma L_r} \right) i_{dr} + (\omega_s - \omega_r) i_{qr} \\ &\quad - \frac{\omega_r L_m \psi_s}{\sigma L_r L_s} + \left(-\frac{L_m}{\sigma L_s L_r} \right) u_{ds} + \frac{1}{\sigma L_r} u_{dr} \\ \dot{i}_{qr} &= -(\omega_s - \omega_r) i_{dr} + \left(-\frac{R_s L_m^2}{\sigma L_s^2 L_r} - \frac{R_r}{\sigma L_r} \right) i_{qr} \\ &\quad + \frac{R_s L_m \psi_s}{\sigma L_s^2 L_r} + \left(-\frac{L_m}{\sigma L_s L_r} \right) u_{qs} + \frac{1}{\sigma L_r} u_{qr} \\ \dot{\omega}_r &= \left(\frac{n_p}{J} \right) \left(\frac{3n_p L_m \psi_s}{2L_s} i_{dr} - \frac{D}{n_p} \omega_r - T_L \right) \end{aligned} \quad (9)$$

We now rewrite (9) by introducing the following state variables:

$$x_1 = i_{dr}, \quad x_2 = i_{qr}, \quad x_3 = \omega_r.$$

Let the parameters $a_1, a_2, \dots, a_8$ be defined accordingly. The resulting nonlinear system can then be expressed in the compact state-space form

$$\begin{cases} \dot{x}_1 = a_1 x_1 + (\omega_s - x_3) x_2 + a_2 x_3 + (a_3 u_{ds} + a_4 u_{dr}), \\ \dot{x}_2 = -(\omega_s - x_3) x_1 + a_1 x_2 + a_5 + (a_3 u_{qs} + a_4 u_{qr}), \\ \dot{x}_3 = a_6 x_1 - a_7 x_3 - a_8. \end{cases} \quad (10)$$

The coefficients appearing in (10) are given by

$$\begin{aligned} a_1 &= -\frac{R_s L_m^2}{\sigma L_s^2 L_r} - \frac{R_r}{\sigma L_r}, \quad a_2 = -\frac{L_m \psi_s}{\sigma L_r L_s}, \quad a_3 = -\frac{L_m}{\sigma L_s L_r}, \\ a_4 &= \frac{1}{\sigma L_r}, \quad a_5 = \frac{R_s L_m \psi_s}{\sigma L_s^2 L_r}, \quad a_6 = \frac{3n_p^2 L_m \psi_s}{2J L_s}, \\ a_7 &= \frac{D}{J}, \quad a_8 = \frac{n_p}{J} T_L, \end{aligned}$$

with $\sigma = 1 - L_m^2/(L_s L_r)$. The stator flux is aligned with the q axis, so that $u_{ds} = -\omega_s \psi_s$ and $u_{qs} = 0$.

TABLE I
MAIN PARAMETERS OF THE CHAOTIC DFIG SYSTEM

Parameter	Value	Parameter	Value
R_s	$5.0536 \times 10^{-5} \Omega$	n_p	2
R_r	$1.8781 \times 10^{-4} \Omega$	J	$31.18 \text{ kg} \cdot \text{m}^2$
L_s	0.0833 H	D	$71.85 \text{ N} \cdot \text{m} \cdot \text{s/rad}$
L_r	0.0844 H	T_L	$-11337 \text{ N} \cdot \text{m}$
L_m	0.0811 H	u_{dr}	2.539 V
σ	0.0645	u_{qr}	-2.936 V
ω_s	$100\pi \text{ rad/s}$	u_{ds}	-563.4 V
ψ_s	1.7933 Wb	u_{qs}	0 V

The parameters listed in Table I are physically consistent: $L_s, L_r > L_m$ and $\sigma = 0.0645 \in (0, 1)$. They give $a_1 = -0.0433$, $a_2 = -320.83$, $a_3 = -178.91$, $a_4 = 183.76$, $a_5 = 0.1946$, $a_6 = 0.3360$, $a_7 = 2.3044$, $a_8 = -727.20$. The constant terms of (10) are therefore $a_3 u_{ds} + a_4 u_{dr} = 1.0126 \times 10^5$ and $a_5 + a_3 u_{qs} + a_4 u_{qr} = -539.32$.

System (10) has three equilibrium points, all unstable:

- $E_1 = (-64.94, -379.12, 306.11)$, a saddle-focus with eigenvalues $1.462 \pm 7.673j$ and -5.315 ;
- $E_2 = (-9.97, -12442.5, 314.12)$, a saddle with eigenvalues $-65.00, 62.65$ and -0.043 ;
- $E_3 = (55.50, -272.00, 323.67)$, a saddle-focus with eigenvalues $0.547 \pm 10.511j$ and -3.486 .

The divergence of the vector field is constant and negative, $\nabla \cdot f = 2a_1 - a_7 = -2.391$, so the system is dissipative.

With these parameters, the dynamical system exhibits complex nonlinear behavior. The corresponding chaotic attractors generated from the initial condition $x(0) = (0.3, 0.6, 3.6)^T$ are illustrated in Fig. 1.

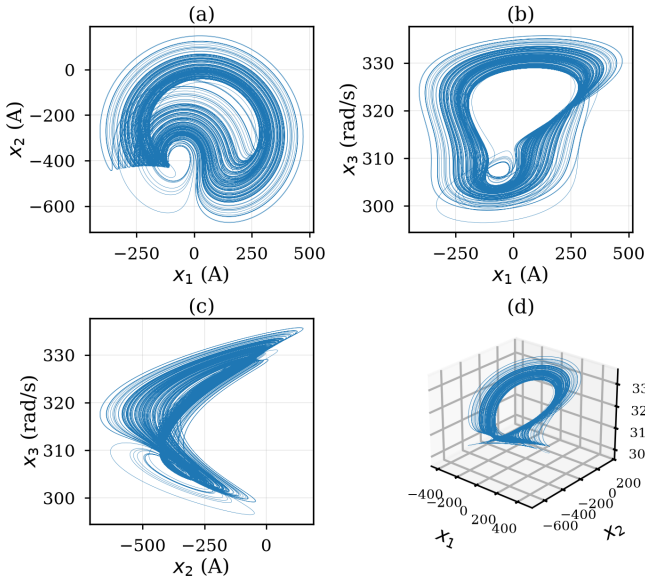


Fig. 1. The chaotic attractors: a) The projection of the chaotic attractor in plan $x_1 - x_2$; b) the projection of the chaotic attractor in plan $x_1 - x_3$; c) the projection of the chaotic attractor in plan $x_2 - x_3$; d) 3D trajectory of the system.

The phase portraits presented in Fig. 1 illustrate the complex nonlinear behavior of the DFIG system. The trajectory winds around the two saddle-foci E_1 and E_3 . The rotor speed remains positive and close to synchronism ($296 \leq \omega_r \leq 336 \text{ rad/s}$), while the rotor currents oscillate irregularly between about -670 and $+470 \text{ A}$. The Lyapunov exponents, computed with the QR-based variational method, are $(0.202, -0.010, -2.582)$. Their sum equals the divergence -2.391 , and the Kaplan-Yorke dimension is 2.08 . The positive largest Lyapunov exponent confirms the presence of chaotic dynamics. The three-dimensional attractor further highlights the bounded yet aperiodic evolution of the system states, which is a typical signature of chaos.

IV. REFERENCE TRAJECTORY GENERATION AND TRACKING ERROR FORMULATION

A realistic wind excitation is essential for evaluating the dynamic performance of a DFIG-based wind energy conversion system. In this work, the wind speed is modeled as the combination of a mean component, several low-frequency harmonic components, and a stochastic turbulence term, yielding a physically representative turbulent wind signal. The wind-speed model is expressed as

$$v(t) = V_{mean} + \sum_{k=1}^3 A_k \sin(2\pi f_k t) + v_{turb}(t) \quad (11)$$

here V_{mean} denotes the average wind speed, A_k and f_k are the amplitudes and frequencies of the harmonic components, and $v_{turb}(t)$ is a smoothed Gaussian noise process representing atmospheric turbulence.

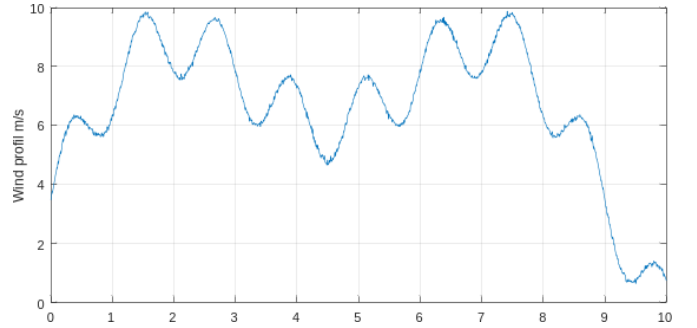


Fig. 2. Wind-speed trajectory.

Fig. 2 shows the generated wind-speed profile used to excite the DFIG system. The signal combines deterministic low-frequency variations with stochastic turbulence, providing a realistic representation of wind conditions. This complex excitation plays a key role in revealing the nonlinear and potentially chaotic behavior of the system.

Based on this profile, the aerodynamic power reference is computed from the standard wind-power expression

$$P_{ref}(t) = \frac{1}{2} \rho \pi R^2 C_p v^3(t) \quad (12)$$

where ρ is the air density, R the turbine radius, and C_p the optimal power coefficient. Under stator-flux orientation, the

active power is primarily governed by the q -axis rotor current, leading to the reference

$$i_{q,ref}(t) = \frac{2}{3} \frac{P_{ref}(t)}{V_s} \frac{L_s}{L_m} \quad (13)$$

The corresponding measured active power is

$$P_m(t) = \frac{3}{2} \frac{L_m}{L_s} V_s i_q(t) \quad (14)$$

The reference for the d -axis rotor current is derived from the stator reactive-power expression under stator-flux orientation,

$$i_{d,ref}(t) = -\frac{2L_s}{3L_m V_s} \left(Q_s(t) - \frac{3V_s^2}{2\omega_s L_s} \right) \quad (15)$$

and, by imposing $Q_s(t) = 0$, simplifies to the constant operating point

$$i_{d,ref} = \frac{V_s}{\omega_s L_m} \quad (16)$$

which ensures unity power-factor operation.

The reference rotor speed is obtained from the optimal tip-speed ratio, giving

$$\omega_{r,ref}(t) = \frac{G \lambda_{opt}}{R} v(t) \quad (17)$$

The complete reference trajectory is therefore

$$x_r(t) = \begin{bmatrix} i_{d,ref}(t) \\ i_{q,ref}(t) \\ \omega_{r,ref}(t) \end{bmatrix},$$

and the corresponding tracking errors are defined as

$$e(t) = x(t) - x_r(t), \quad e_i(t) = x_i(t) - x_{r,i}(t), \quad i = 1, 2, 3.$$

A common practice in addressing the control problem of wind turbines is to use a linearization approach. However, due to the stochastic operating conditions and the inevitable uncertainties inherent in the system, such control methods come at the price of poor system performance and low reliability. Hence the need for nonlinear and robust control to take into account these control problems.

V. HYBRID CONTROL OF A NOVEL CHAOTIC DFIG-BASED SYSTEM

The controlled system takes the form

$$\begin{aligned} \dot{x}_1 &= a_1 x_1 + (\omega_s - x_3) x_2 + a_2 x_3 + a_3 u_{ds} + a_4 u_1, \\ \dot{x}_2 &= -(\omega_s - x_3) x_1 + a_1 x_2 + a_5 + a_3 u_{qs} + a_4 u_2, \\ \dot{x}_3 &= a_6 x_1 - a_7 x_3 - a_8, \end{aligned} \quad (18)$$

where $x = [x_1, x_2, x_3]^T = [i_{dr}, i_{qr}, \omega_r]^T$ and the two control inputs $u_1 = u_{dr}$ and $u_2 = u_{qr}$ are the rotor voltages imposed by the rotor-side converter. The speed equation has no direct input: ω_r is controlled through the electromagnetic torque, i.e., through i_{dr} . With the parameters of Table I, (18) reads

$$\begin{aligned} \dot{x}_1 &= -0.0433 x_1 + (\omega_s - x_3) x_2 - 320.83 x_3 \\ &\quad + 1.0079 \times 10^5 + 183.76 u_1, \\ \dot{x}_2 &= -(\omega_s - x_3) x_1 - 0.0433 x_2 + 0.1946 + 183.76 u_2, \\ \dot{x}_3 &= 0.3360 x_1 - 2.3044 x_3 + 727.20. \end{aligned}$$

In open loop ($u_1 = 2.539$ V, $u_2 = -2.936$ V), (18) reduces to the chaotic system (10). The objective is to force the nonlinear chaotic dynamics to follow a desired trajectory $x_d(t)$, constructed from wind-profile-based references for i_{dr} , i_{qr} , and ω_r . The tracking error is defined as $e(t) = x(t) - x_d(t)$, and its stabilization ensures both chaos suppression and accurate MPPT behavior under turbulent wind fluctuations.

The overall control structure implementing this hybrid strategy combining Model Predictive Control with nonlinear feedback compensation is illustrated in Fig. 3, where the proposed controller interacts with the turbine dynamics and the MPPT block to achieve robust trajectory tracking under variable wind conditions.

This hybrid framework harnesses the strengths of both techniques to overcome their individual limitations and improve overall system performance. Predictive control [25]–[30] is particularly valued for its ease of configuration and implementation, which makes it a widely adopted technique for controlling chaotic systems. Its efficiency, robustness, and straightforward design process enable effective management of complex dynamics with relatively low computational effort. Feedback control excels in stabilizing inherently unstable and unpredictable chaotic dynamics by continuously adjusting the control input in real time based on the system's current state. Additionally, feedback control [31], [32] offers significant adaptability, allowing synchronization or steering of chaotic trajectories toward desired states, which is crucial for practical applications such as cryptography, robotics, and biological systems. By integrating these two approaches, the proposed hybrid control strategy capitalizes on the simplicity and robustness of predictive control alongside the real-time adaptability and energy efficiency of feedback control, thereby providing a more comprehensive solution for managing chaotic systems.

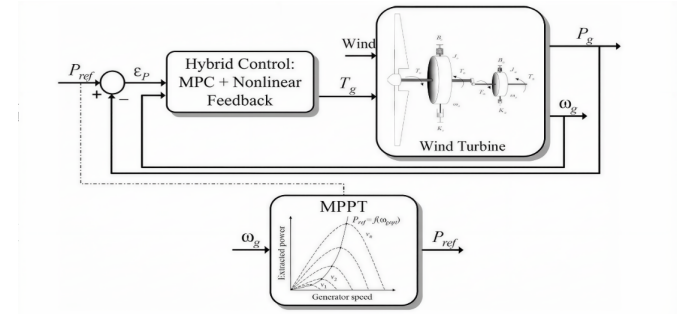


Fig. 3. Proposed hybrid MPC–nonlinear feedback control architecture for DFIG-based wind turbines.

Theorem 1: The system (18) is asymptotically stable under the control law:

$$U = K_1 (\dot{x} - x - \dot{x}_{ref}) + K_2 (x - x_{ref}), \quad (19)$$

if and only if there exists a positive definite symmetric matrix $X = X^T = P > 0$ and a matrix Y for any symmetric matrix $Q = Q^T > 0$, satisfying the inequality:

$$\begin{bmatrix} B & I \\ I & -Q^{-1} \end{bmatrix} < 0$$

with

$$B = Df(x_f)^T P + PDf(x_f) + Df(x_f)^T K_1^T P + PK_1 Df(x_f) + PK_2 + K_2^T P - (PK_2 + K_2^T P)$$

Proof: By linearizing the nonlinear system around the equilibrium x_f and using the first-order Taylor expansion of $f(x)$, we obtain

$$\dot{\delta x}(t) = A\delta x(t) + \delta u(t), \quad A := Df(x_f). \quad (20)$$

With the control law (19) and in deviation coordinates about x_f (so that $x_{ref} = \text{constant}$), the linearized closed loop is

$$\begin{aligned} \delta \dot{x}(t) &= A\delta x(t) + K_1(A\delta x(t) - \delta x(t)) + K_2\delta x(t) \\ &= (A + K_1A - K_1 + K_2)\delta x(t) =: A_{ce}\delta x(t). \end{aligned}$$

We consider the Lyapunov function

$$V(x(t)) = x^T(t) P x(t), \quad P = P^T > 0. \quad (21)$$

Its derivative along the trajectories of V is

$$\begin{aligned} \dot{V}(x(t)) &= \dot{x}^T(t) P x(t) + x^T(t) P \dot{x}(t) \\ &= x^T(t) (A_{ce}^T P + P A_{ce}) x(t) \\ &= x^T(t) \left(A^T P + P A + A^T K_1^T P + P K_1 A - (K_1^T P + P K_1) + (P K_2 + K_2^T P) \right) x(t). \end{aligned}$$

Introduce the standard change of variables

$$Y_1 = P K_1, \quad Y_2 = P K_2,$$

to eliminate the bilinear terms. The above then becomes

$$\begin{aligned} \dot{V}(x(t)) &= x^T(t) B x(t), \\ B &:= A^T P + P A + A^T Y_1^T + Y_1 A + Y_2 + Y_2^T - (Y_1 + Y_1^T). \end{aligned}$$

If the following LMI holds for some $Q = Q^T > 0$:

$$B + Q < 0,$$

then

$$\dot{V}(x(t)) \leq -x^T(t) Q x(t) < 0 \quad \text{for all } x(t) \neq 0,$$

which proves that the closed-loop system is asymptotically stable.

From the Schur complement, this condition is equivalent to the block matrix condition

$$S = \begin{pmatrix} B & I \\ I & -Q^{-1} \end{pmatrix} < 0.$$

The theorem guarantees stability for a general nonlinear control structure. In the specific chaotic DFIGN application, the control input is implemented as

$$u = -f(x) + \dot{x}_d - K(x - x_d) \quad (22)$$

where $K = \text{diag}(k_1, k_2, k_3)$ is a stabilizing diagonal gain matrix selected to ensure rapid convergence of the tracking error.

VI. NUMERICAL SIMULATIONS

In this section, numerical simulations are presented to demonstrate and verify the performance of the proposed approach. To stabilize the chaotic system (10) to the equilibrium point, the system is rewritten as:

$$\dot{x}(t) = Ax(t) + f(x(t)) + u \quad (23)$$

where

$$A = \begin{bmatrix} a_1 & (w_s - x_3) & -x_2 - a_2 \\ -(w_s - x_3) & a_5 & x_1 \\ a_8 & 0 & -a_9 \end{bmatrix}$$

and

$$f(x(t)) = \begin{bmatrix} -x_2x_3 + x_3u_{ds} + a_4u_{dr} \\ -x_3x_1 + x_5 + x_3u_{qs} + a_4u_{qr} \\ -a_8 \end{bmatrix}$$

The system (10) has a single equilibrium point, and the Jacobian matrix around this equilibrium point is

$$J = \begin{bmatrix} 0.0231 & 720.2775 & 23.6504 \\ -720.2775 & 0.0231 & -0.00075366 \\ 48579 & 0 & -0.1 \end{bmatrix}$$

Then, we apply the control input $u(t)$ proposed in Theorem 1. The feedback gains are determined by solving the LMIs from Theorem 1 using MATLAB 2024a with the appropriate toolbox. The obtained results are:

$$k_1 = \begin{bmatrix} -73.1801 & 38.6679 & -25.9740 \\ 38.6679 & -74.4617 & 21.9579 \\ -7.9740 & 3.9579 & -21.8599 \end{bmatrix}$$

$$k_2 = \begin{bmatrix} -570.4974 & 36.6083 & 19.3025 \\ 36.6083 & -393.2214 & -572.8579 \\ 19.3025 & -224.8579 & -192.1064 \end{bmatrix}$$

At $t = 5$ s, the control input is introduced to the system, causing a notable change in its dynamic response.

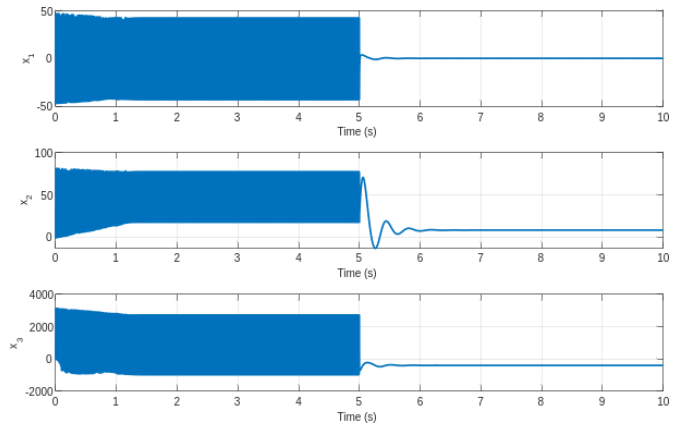


Fig. 4. Time evolution of the state variables before and after control activation.

Fig. 4 shows the time evolution of the system states before and after control activation. Prior to $t = 5$ s, the system exhibits chaotic oscillations. Once the control is applied, the

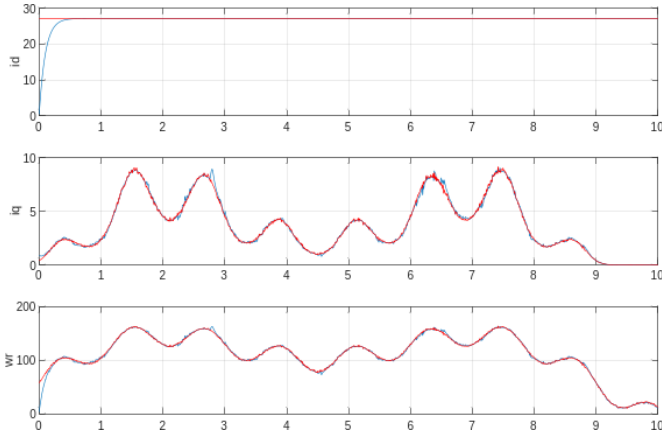


Fig. 5. Tracking performance of the state variables i_{dr} , i_{qr} , and ω_r under the proposed hybrid control scheme.

states rapidly converge toward the equilibrium point, demonstrating the effectiveness of the proposed stabilization strategy.

Fig. 5 demonstrates the tracking capability of the proposed controller. The state variables closely follow their respective reference trajectories despite the presence of wind disturbances and system nonlinearities, indicating high tracking accuracy and robustness, and Fig. 6 presents the active power response of the DFIG system. The measured power closely follows the reference signal derived from MPPT, confirming the effectiveness of the proposed control strategy in maximizing energy extraction while maintaining system stability.

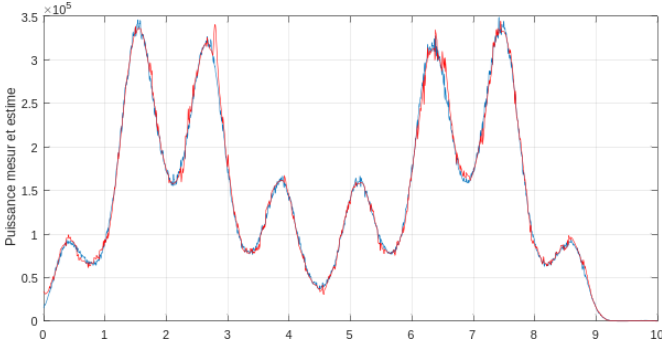


Fig. 6. Active power tracking performance: comparison between measured power P_m and reference power P_{ref} under realistic turbulent wind profiles.

VII. CONCLUSION

This paper investigated the nonlinear dynamics of a DFIG-based wind energy conversion system, highlighting the emergence of complex and chaotic behaviors under turbulent wind conditions, parameter variations, and adverse operating scenarios. Using phase portraits, Lyapunov exponents, and bifurcation analysis, the intrinsic sources of instability arising from the electromechanical coupling of the DFIG were identified. The results show that unmitigated chaotic dynamics can severely degrade power quality, MPPT performance, and overall system stability. To overcome these issues, a hybrid

control strategy combining Model Predictive Control with nonlinear state-feedback compensation was proposed. The controller successfully suppresses chaotic oscillations, ensures accurate reference tracking, and demonstrates strong robustness to uncertainties and wind variability. Simulation results confirm fast convergence of key system variables toward their desired values, even under highly stochastic wind conditions. Overall, the proposed approach significantly improves the stability, reliability, and performance of DFIG-based wind turbines, emphasizing the role of advanced nonlinear control techniques in modern renewable energy systems.

CONFLICT OF INTEREST

The authors declare there are no conflicts of interest.

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